

SCHOOL OF SCIENCE & TECHNOLOGY:
DEPARTMENT OF DATA SCIENCE & ANALYTICS
MTH 2215: DISCRETE MATHEMATICS

MODEL QUESTIONS

1. Write True or False against each of the following statements:

(a) A course in discrete mathematics provides the abstract mathematical structures used to represent discrete objects and relationships between objects.

(b) Discrete structures include sets, permutations, relations, graphs, trees, and finite state machines.

2. What is discrete mathematics?

3. State the kinds of problems solved using discrete mathematics.

4. State the important reasons for studying discrete mathematics.

5. Let $A = \{a, b, c, d, g\}$, $B = \{b, c, d, e, f\}$ and $C = \{b, c, e, g, h\}$. Find $A \cup (B \cap C)$ and $(A \cup B) \cap (A \cap C)$. Show this result by Venn diagram.

6. Let $A = \{a, \dots, z, 0, \dots, 9\}$, and $B = \{0, \dots, 9\}$. Then find (i) $A - B$ and (ii) $B - A$

7. Let $U = \{a, \dots, z\}$. Find the complements of the sets $A = \{a, e, i, o, u\}$ and $B = \{a, c, d, e, \dots, w\}$.

8. Let $A = \{a, b, x, y, z\}$, $B = \{c, d, e, x, y, z\}$, and $U = \{a, b, c, d, e, w, x, y, z\}$. Find (i) $(A \cup B)'$ and $A' \cap B'$.

9. Let $A = \{a, \dots, z, 0, \dots, 9\}$ and $B = \{0, \dots, 9, +, -, *, /\}$. Then find (i) $A - B$; (ii) $B - A$ and (iii) $A \oplus B$ (The symmetric difference of A and B).

(iv) Display the symmetric difference of A and B pictorially by Venn diagram in three three different cases.

10. Draw the table for Set operatin Vs Logic Operation.

11. Prove that $(A \cup B)' = A' \cap B'$ and represent the result by Venn diagram.

12. Using the truth table, prove that $(A \cup B)' = A' \cap B'$.

13. Using set laws, verify that $(X - Y) - Z = X - (Y - Z)$.

14. Simplify the set expression $(A \cap B') \cup (A' \cap B) \cup (A' \cap B')$.

15. $A = \{a, b\}$ and $B = \{x, y, z\}$. Then find the Cartesian products (i) $A \times B$; (ii) $B \times A$; (iii) A^2 ; (iv) Is $A \times B = B \times A$?

16. Let $A = \{x\}$, $B = \{y, z\}$, and $C = \{1, 2, 3\}$. Then find the Cartesian product $A \times B \times C$

17. Write the definition of partition & give an example.

18. State and prove the Inclusion Exclusion Principle.

19. State the addition Principle.

20. Find the number of positive integers ≤ 300 and divisible by 2 or 3.

21. A survey of 100 students shows that of the three ice cream flavours vanilla, chocolate, and strawberry, 50 students like vanilla, 43 like chocolate, 28 like strawberry, 13 like vanilla and chocolate, 11 like chocolate and strawberry, 12 like strawberry and vanilla, and 5 like all of them. Find the number of students surveyed who like each of the flavours:

(i) Chocolate but not strawberry.

(ii) Chocolate and strawberry, but not vanilla.

(iii) Vanilla or chocolate, but not strawberry.

22. Write the definition of proposition & give two examples. (2 Marks)

23. Write the definition of conjunction & give an example. (2 Marks)

24. Write the truth table for conjunction. (2 Marks)

25. Write the definition of disjunction & give an example. (2 Marks)

26. Write the truth table for disjunction. (2 Marks)

27. Write the definition of negation & give an example (2 Marks)

28. Write the truth table for negation. (2 Marks)

29. Evaluate each Boolean expression, where $a = 3$, $b = 5$, and $c = 6$.

(i) $[\sim (a > b)] \wedge (b < c)$ (ii) $\sim[(a \leq b) \vee (b > c)]$

30. Write the definition of implication statement & give an example (2 Marks)

31. Write the truth table for implication statements. (2 Marks)

32. Write about the converse, inverse and contrapositive of implication statement & explain the concepts by an example. (4 marks)

33. Construct a truth table for $(p \rightarrow q) \wedge (q \rightarrow p)$. (4 Marks)
34. Write the definition of bicondition statement and give an example. (2 Marks)
35. Write the truth table for bicondition statement. (2 Marks)
36. Construct a truth table for $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$. (2 Marks)
37. Write about what do you mean by tautology, contradiction and contingency.
38. Draw the switing network figure for the following symbolic representation:
 $(A \wedge B') \vee B \vee (A' \wedge C)$
39. Using truth table, show that $p \rightarrow q \equiv \sim q \rightarrow \sim p$; that is, an implication is logically equivalent to its contrapositive.
40. Using truth table, verify that $\sim (p \wedge q) \equiv \sim p \vee \sim q$.
41. Determine whether or not the statement $x \leftarrow x + 1$ will be executed in the following sequence of statements:
 $a \leftarrow 7; b \leftarrow 4$
 if $\sim[(a < b) \vee (b \geq 5)]$ then
 $x \leftarrow x + 1$
42. Using the laws of logic simplify the boolean expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$
- 43. Write the product rule (2 Marks)**
- 44. A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees? (2 Marks)**
- 45. The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently? (2 Marks)**
- 46. There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there? (2 Marks)**
- 47. How many different bit strings of length seven are there? (2 Marks)**
- 48. How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits? (2 Marks)**
- 49. How many functions are there from a set with m elements to a set with n elements? (2 Marks)**

50. How many one-to-one functions are there from a set with m elements to one with n elements? (2 marks)

51. Write the sum rule. (2 Marks)

52. Suppose that either a member of mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both faculty member and a student? (2 Marks)

53. Write True or False against following statement: A course in discrete mathematics provides the abstract mathematical structures used to represent discrete objects and relationships between objects. (1 Mark)

54 . Write True or False against the following statement: Discrete structures include sets, permutations, relations, graphs, trees, and finite state machines. (1 Mark)

55. The n th term of a geometric sequence is $u_n = \left(\frac{2}{3}\right)^n$. Answer the following:

What are the first three terms? (1 Mark)

(a) $\frac{2}{3}, \frac{4}{9}, \frac{8}{27}$ (b) $\frac{7}{3}, \frac{8}{9}, \frac{12}{27}$ (c) $\frac{5}{3}, \frac{7}{9}, \frac{2}{27}$ (d) None

56. The n th term of a geometric sequence is $u_n = \left(\frac{2}{3}\right)^n$. Answer the following:

What is the value of S_5 , the sum of the first five terms? (1 Mark)

(a) $\frac{\frac{2}{3}[1-(\frac{2}{3})^5]}{1-\frac{2}{3}}$ (b) $\frac{\frac{2}{3}[-(\frac{2}{3})^5]}{-\frac{2}{3}}$ (c) $\frac{\frac{2}{3}[1-(\frac{2}{3})^4]}{1+\frac{2}{5}}$ (d) None

57. Let S be the set of letters in the English alphabet. Then $|S| = ?$ (1 Mark)

(a) $|S| = 26$ (b) $|S| = 27$ (c) $|S| = 25$ (d) None

58. Let A and B be finite disjoint sets. Then $|A \cup B| = ?$ (1 Mark)

(a) $|A| + |B|$ (b) $|A|$ (c) $|B|$ (d) None

59. How many number of positive integers less than ($\leq$) 300 and divided by 2 or 3 are there? (1 Mark)

(a) 300 (b) 200 (c) 100 (d) None

60. What is the negation of the proposition: "Michael's PC runs Linux"? (1 Mark)

(a) "Michael's PC does not run Linux" (b) "Michael's PC does run Linux"
(c) "Michael's PC runs Linux" (d) None

61. If 2, 6, 18, ..., 1458 is a geometric sequence, answer the following:

What is the n th term u_n ? (1 Mark)

(a) $2(3)^{n-1}$ (b) $2(3)^n$ (c) $(3)^{n-1}$ (d) None

62. If 2, 6, 18, ..., 1458 is a geometric sequence, answer the following:

What are the number of terms (the value of n), in the sequence? (1 Mark)

(a) 7 (b) 6 (c) 5 (d) None

63. Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2$, $f(b) = 1$, $f(c) = 4$, $f(d) = 1$ and $f(e) = 1$. Then what is the image of the subset $S = \{b, c, d\}$? (1 Mark)

(a) The set $f(S) = \{1, 4\}$ (b) The set $f(S) = \{1, 3\}$ (c) The set $f(S) = \{1, 2\}$ (d) None

64. Let $f(x) = x^2$ and $g(x) = \sqrt{x-1}$, where $\text{dom}(f) = (-\infty, \infty)$ and $\text{dom}(g) = [1, \infty)$. Then what is $(f+g)(x)$ and what is $(f \circ g)(x)$? (1 Mark)

(a) $x^2 + \sqrt{x-1}$ and $x^2\sqrt{x-1}$ (b) $\sqrt{x-1}$ and 1 (c) $1+\sqrt{x-1}$ and $\sqrt{x-1}$ (d) None

65. Three terms in a geometric sequence are $x-3$, x , $3x+4$, (where x is a real number).

What are the two possible values of x ? (1 Mark)

(a) $-\frac{3}{2}$ or 4 (b) $-\frac{5}{2}$ or 5 (c) $-\frac{7}{2}$ or 7 (d) None

66. Suppose that $\sum_{n=0}^{\infty} (2x-3)^n = 1 + (2x-3) + (2x-3)^2 + (2x-3)^3 + \dots$ is a geometric series. Answer the following: What is the sum to infinity (in terms of x)? (1 Mark)

(a) $\frac{1}{4-2x}$ (b) $\frac{1}{7-3x}$ (c) $\frac{1}{5-7x}$ (d) None

67. Suppose that $\sum_{n=0}^{\infty} (2x-3)^n = 1 + (2x-3) + (2x-3)^2 + (2x-3)^3 + \dots$ is a geometric series. Answer the following: If the sum to infinity is $\frac{5}{4}$, what is the value of x ? (1 Mark)

(a) $\frac{8}{5}$ (b) $\frac{7}{5}$ (c) $\frac{6}{5}$ (d) None

68. Write true or false for the following statement:

A pseudograph is with undirected multiple edges and loops. (1 Mark)

69. Write true or false for the following statement:

A simple directed graph is with directed edges, no multiple edges and no loops. (1 Mark)

70. Write true or false for the following statement:

A directed multigraph is with directed multiple edges and loops. (1 Mark)

71. If $k+2$, $2k+3$, $5k-2$ are three consecutive terms in an arithmetic sequence, then, what is the value of k ? (1 Mark)

(a) 3 (b) 2 (c) 1 (d) None

72. What is the sum of the arithmetic series $5 + 8 + 11 + \dots + 65$? (1 Mark)

(a) 735 (b) 730 (c) 725 (d) None

73. Write true or false for the following statement:

A mixed graph is with directed and undirected multiple edges and loops. (1 Mark)

74. Write true or false for the following statement:

An undirected graph has an even number of vertices of odd degree. (1 Mark)

75. The sum of the first n terms of an arithmetic series is given by $S_n = 4n - n^2$. Answer the following: What is the value of u_n (n -th term) ? (1 Mark)

(a) $5 - 2n$ (b) $3 - 2n$ (c) $4 - n$ (d) None

76. The sum of the first n terms of an arithmetic series is given by $S_n = 4n - n^2$. Answer the following: What is the value of u_{18} ? (1 Mark)

(a) -31 (b) -30 (c) -32 (d) None

77. The sum of the first n terms of an arithmetic series is given by $S_n = 4n - n^2$. Answer the following: If $S_n = -60$, what is the value of n ? (1 Mark)

(a) 10 (b) 9 (c) 8 (d) None

78. Write true or false for the following statement:

The cyclic graph C_6 is bipartite and K_3 is not bipartite. (1 Mark)

79. Write true or false for the following statement:

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no adjacent vertices are assigned the same color. (1 Mark)

80. How many edges are there in a graph with 10 vertices each of degree six? (1 Mark)

(a) 30 (b) 25 (c) 20 (d) None

81. Write true or false for the following statement:

A simple graph is with undirected edges, no multiple edges and no loops. (1 Mark)

82. Write true or false for the following statement:

A multi graph is with undirected multiple edges and no loops. (1 Mark)